FIFA WORLD CUP PERFORMANCE DASHBOARD (2002–2018)

BY

UJAM PRECIOUS MMESOMA

JULY 19, 2026

TOOLS USED: MICROSOFT POWER BI DESKTOP, POWER QUERY, DAX

DATASET: FIFA WORLD CUP TEAM PERFORMANCE DATASET (2002–2018)

Table of Contents

EXECUTIVE SUMMARY

This report presents an interactive Power BI dashboard developed to analyze the performance of national football teams across the FIFA World Cup tournaments from 2002 to 2018. The objective of the dashboard is to provide decision-makers with an intuitive platform for monitoring team performance using key performance indicators such as wins, losses, draws, goals scored, goals conceded, goal difference, points, and win rate.

The solution was built using a star schema consisting of one fact table and supporting dimension tables for Date and Country. Power Query was used to clean and transform the dataset, while DAX measures were developed to calculate business metrics including Win Rate, Average Goals per Game, Average Points, Draw Rate, Loss Rate and Goal Difference.

Three interactive report pages were designed. The Executive Overview presents high-level KPIs and ranking visuals. The Performance Analysis page provides comparative analysis of team statistics and tournament trends. The Team Details page uses drill-through functionality to allow users analyze the historical performance of individual teams.

The dashboard demonstrates how business intelligence techniques can transform raw tournament data into meaningful insights. It enables users to identify top-performing teams, evaluate performance trends over time and compare teams using multiple performance metrics. The solution also incorporates interactive filtering, drill-through navigation and dynamic report titles to improve usability and user experience.

BUSINESS CONTEXT & PROBLEM STATEMENT

International football competitions generate large volumes of performance statistics every tournament. However, analyzing this information manually makes it difficult to identify long-term trends, compare teams objectively and support strategic decision-making.

The objective of this project is to develop a Power BI dashboard that transforms historical FIFA World Cup data into an interactive reporting solution capable of answering key business questions regarding team performance.

The dashboard is intended to support analysts, football researchers, coaches and sports administrators by providing quick access to important performance metrics and tournament trends. Users can filter by tournament year or team, compare performance across competitions and drill through into individual team reports.

The primary objectives include:

- Evaluate overall team performance across World Cup tournaments.
- Compare teams using wins, goals scored, goals conceded and total points.
- Identify the highest-performing teams.
- Analyze tournament performance trends over time.
- Provide detailed analysis for individual teams using drill-through functionality.

DATA MODEL SUMMARY

The solution follows a Star Schema data model consisting of one fact table and two dimension tables.

Fact Table:

- FactFIFA
- Stores tournament statistics including Games Played, Wins, Draws, Losses, Goals For, Goals Against, Goal Difference, Points, Position and Tournament Year.

Dimension Tables:

DimCountry

- Contains unique country names used for filtering and categorization.

DimDate

- Contains Date, Year, Quarter, Month, Month Number and Day fields to support time-based analysis.

Relationships were created between the fact table and each dimension table using one-to-many relationships with single-direction filtering. This design improves query performance and follows Power BI modelling best practices. A detailed description of the dataset, including field definitions and metadata, is provided in the accompanying **Data Dictionary Deliverable**.

DATA QUALITY & TRANSFORMATION NOTES

Several data quality issues were identified during data preparation.

Power Query transformations included:

- Removed duplicate records.
- Corrected inconsistent data types.
- Converted text columns into numeric values.
- Replaced invalid characters such as en-dash symbols in numeric fields.
- Created Tournament Year from the Date field.
- Built a dedicated Date table for time intelligence.
- Renamed fields for consistency.
- Verified row counts after loading.

These transformations improved data accuracy and ensured that all DAX calculations produced reliable results.

KEY BUSINESS QUESTIONS

What is the trend of total goals scored over the last five tournaments (2002–2018)?

The dashboard shows that total goals scored declined between the 2002 and 2010 FIFA World Cups before increasing significantly in 2014. Although there was a slight decrease in 2018, the tournament remained one of the highest-scoring editions during the period analyzed. This indicates an overall upward trend in offensive performance after 2010.

In which tournament did France achieve its highest points, goals scored, and wins?

By filtering the dashboard to France, the analysis identifies the tournament year in which France recorded its highest total points, goals scored, and number of wins (2018). This corresponds to France's strongest performance within the analyzed period.

What is the trend of games won and lost by France?

The dashboard indicates that France's wins and losses varied across tournaments. The team's most successful tournament is characterized by a higher number of wins and fewer losses, demonstrating a strong relationship between match outcomes and tournament success.

Which country has the highest and lowest goal difference?

The dashboard identifies Germany with the highest positive goal difference of 22, as the strongest offensive and defensive performer over the selected tournaments. Conversely, Saudi Arabia, the team with the lowest goal difference (-22) conceded substantially more goals than it scored, indicating comparatively weaker performance.

How did Nigeria perform during the analyzed tournaments?

Filtering the dashboard to Nigeria shows its performance across tournaments based on matches played, wins, losses, goals scored, goals conceded, and total points. The analysis highlights periods of improvement and decline, providing insight into Nigeria's overall tournament performance.

KEY INSIGHTS

Insight 1:

Germany and Brazil consistently ranked among the highest-performing teams based on total points accumulated across tournaments, demonstrating sustained success throughout the study period.

Insight 2:

Tournament goal scoring increased significantly during the 2014 and 2018 FIFA World Cups compared to earlier tournaments, indicating more attacking styles of play.

Insight 3:

Several teams maintained strong win rates despite playing fewer matches, showing higher competitive efficiency than teams with larger numbers of games played.

Insight 4:

Goals scored strongly influenced total points and tournament success, while teams conceding fewer goals generally achieved higher rankings.

Insight 5:

The drill-through analysis revealed that team performance fluctuated considerably across tournament years, highlighting the importance of analyzing performance historically rather than relying on cumulative statistics alone.

RECOMMENDATIONS

Recommendation 1:

Continue monitoring win rate alongside total points because teams with higher win percentages consistently achieve stronger tournament outcomes.

Recommendation 2:

Focus on improving defensive performance since lower goals conceded are associated with higher tournament rankings and greater competitive success.

Recommendation 3:

Use historical tournament trends to evaluate long-term team development instead of relying solely on single tournament performance.

Recommendation 4:

Expand future analysis by incorporating player statistics, possession, shots on target and expected goals (xG) to provide deeper tactical insights.

Recommendation 5:

Automate future dashboard refreshes using scheduled Power BI Service refreshes to ensure reports remain up to date as new tournament data becomes available.

DESIGN RATIONALE

The dashboard was designed using a clean blue colour palette to reflect the FIFA theme while maintaining readability.

Design decisions include:

- KPI cards placed at the far left and top for quick performance overview.
- Consistent colour usage across all report pages.
- Interactive slicers for Team and Tournament Year.
- Drill-through navigation from summary pages to detailed team analysis.
- Dynamic report titles that update based on selected teams.
- Balanced spacing and alignment to reduce visual clutter.
- Rounded containers and grouped visuals to improve dashboard appearance.

The design prioritizes simplicity, consistency and ease of navigation while supporting effective business decision-making.

LIMITATIONS & ASSUMPTIONS

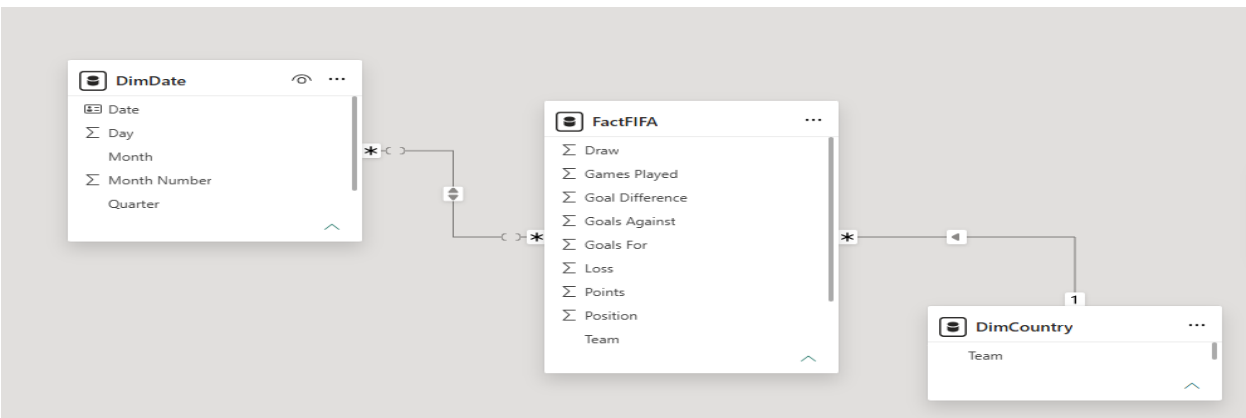
The dashboard has several limitations.

- Analysis covers only FIFA World Cup tournaments between 2002 and 2018.
- Results depend on the accuracy of the source dataset.
- External factors such as injuries, coaching changes and player quality were not included.
- Performance metrics are based solely on historical tournament statistics.
- The dashboard assumes all recorded statistics are complete and accurate.

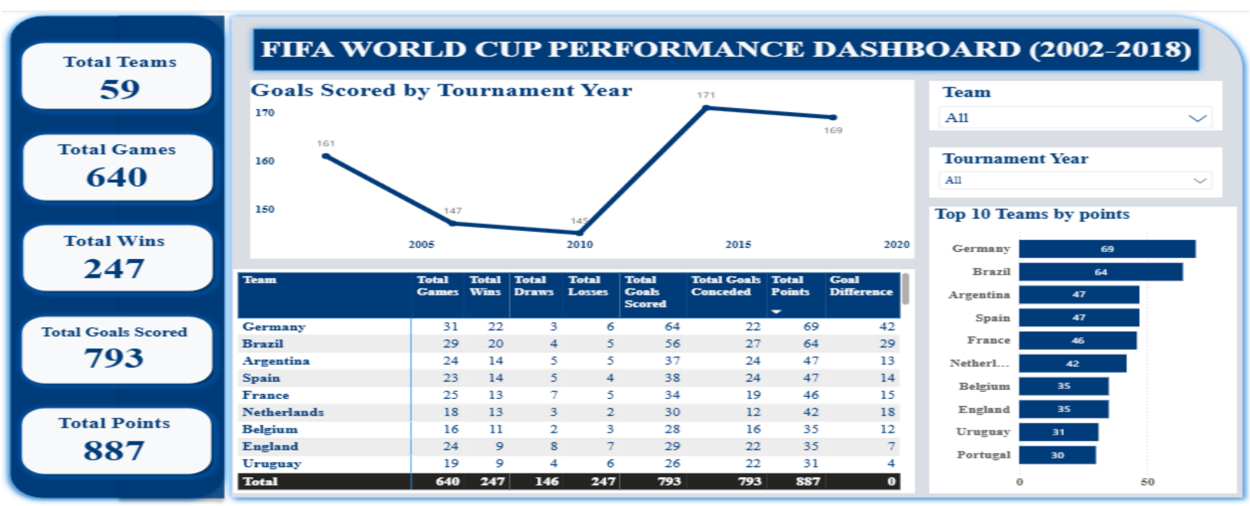
Future improvements could include additional tournaments, player-level statistics, predictive modelling and live data integration.

APPENDIX

Data Model:



Executive Overview:



Nigeria’s performance in the last 5 years: